

Synaptic plasticity as short-term memory

DataForge 2026 · Pathway · Team think loop (solo)
Claim: A fixed-shape synaptic state can process a stream of unbounded length without a new memory slot per token, but similar cues interfere and can retrieve the wrong association.

Design pressure

Transformers retain context with a growing key–value cache. Accurate and observable, yet memory scales with length. A recurring alternative is a fixed-size associative state: write into connections (fast weights), read by cue, accept interference as the capacity cost. This summary covers one mechanism—synaptic / Hebbian short-term memory—because Dragon Hatchling (BDH) makes it architectural.

Mechanism

Teaching write: $\mathbf{W} \leftarrow (\mathbf{1} - \gamma)\mathbf{W} + \eta (\mathbf{k} \otimes \mathbf{v})$. Probe q reads $\hat{\mathbf{v}} = \mathbf{W}q$. Near-colinear keys mix values, but similarity alone is not enough to produce a *wrong* answer: a cue matches its own key at 1.0 and an overlapping key at only $s < 1$, so one competing write lowers confidence without changing the label. Interference wins when overlapping writes accumulate while the original trace decays. In the artifact's preset, $A \rightarrow \text{RED}$ is written once and early and the overlapping key twice and later; the synaptic read for cue A crosses from RED to BLUE at $s^* = u^4 / (u^2 + 1)$ with $u = 1 - \gamma$ (0.39 at default decay), while the KV cache keeps a dedicated slot and stays correct. Write strength η cancels from that expression — a reminder that not every knob is a control variable. The artifact shows truth beside estimate with mixture shares. The simulation is an independent toy—not official BDH weights, and not a Mamba-style SSM (Pathway forbids that classification).

BDH and BDH-CQ

BDH (arXiv:2509.26507): post-Transformer family where reasoning and memory share one fabric; working memory via synaptic plasticity / Hebbian dynamics; sparse non-negative activations and monosemantic synapses reported. Evidence: author-reported paper + educational public toy code.

BDH-CQ (arXiv:2608.09888): same family, different story—demonstrations, latent reasoning without written CoT, recurrent-state adaptation. Comparative, not identical to Hebbian outer-product writes.

Landscape

Approach	Bounded?	Trade-off
Transformer KV	Cache grows	Memory/latency with length
Fast weights / linear attn	Often fixed state	Interference / expressivity
Selective SSMs (Mamba)	Recurrent state	Different bias than BDH
BDH synaptic memory	Associative wiring	Interference; external replication still maturing

Advantage, weakness, maturity

Advantage: constant-shape state follows long streams without a new slot per token; associations are inspectable. **Weakness:** interference; toy omits sparsity/inhibition/scale-free connectivity.

Maturity: active 2022–2026 area; label BDH headline numbers as author-reported unless independently reproduced.

Limitation & next reading

Open question: consolidating useful fast state into durable slow weights. Continue:

arXiv:2509.26507, arXiv:2608.09888, Pathway “From Attention to Synapses,” “Equations of Reasoning,” github.com/pathwaycom/bdh; contrasts LoRA, Mamba, xLSTM. Artifact: [index.html](#) sixty-second test.